

## EVAN J. LONG

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### EDUCATION

**The University of Texas at Austin** – Austin, TX ----- Aug. 2023 – May 2027

*Bachelor of Science in Mechanical Engineering* | GPA: 3.81/4.00 | 93 Course Hours

Relevant Coursework: Machine Elements, Material Analysis, Fluids and Heat Transfer, Robotics Programming and Mechanisms.

**The University of New Mexico - Los Alamos** – Los Alamos, NM ----- May 2021 – May 2023

*Certificate in Robotics* | GPA: 3.86/4.00 | 33 Course Hours

Relevant Coursework: Advanced/Industrial Robotics Projects; Manufacturing Techniques; Hydraulic, Cryogenic, and Vacuum systems.

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### EXPERIENCE

**Hello Maker Studio** – *Mentor / Researcher* ----- Aug 2024 - Present

- Designed and produced various scientific apparatuses for CNS research labs, streamlining their process workflows
- Developed comprehensive training modules for machines, and implemented a feedback loop to improve user experience.
- Mentor students and faculty across colleges and disciplines to refine their designs and manufacturing techniques.

**ASCEND Autonomous Paraglider** – *System Engineer, Engineering Support* ----- Jan. 2024 – Present

- Engineered and produced multiple robust assemblies to autonomously steer the ground vehicle (tube-frame ATV).
- Designed and analyzed a low-cost test stand to verify thrust specs and fine-tune the motor and propeller arrangement.

**EXD Experimental Division Combat Robotics** – *Botfighter* ----- Sep. 2024 – Present

- Designed 3 different PLAntweight battlebots. Optimized all geometries for maximum weight reductions while maintaining 3d-printed structural integrity, prototyped using those parts, and became competition-ready in under a week.
- Current bot (Goomba) fought in an intercollegiate open competition at the University of Houston, and placed 4th in its class.

**Texas Rocket Engineering Lab (TREL)** – *Ground Systems – Vehicle/Stage Test Responsible Engineer* ----- Jan. 2024 – Aug. 2025

- Produced an increased overall project output despite team size reductions, through leading the ground fluids team in developing and building systems to store: gaseous hydrogen, gaseous and liquid oxygen, helium, nitrogen, RP-1, and water.
- Enhanced project planning, documentation, and parts sourcing; fostering collaboration with electronics hardware, software, and structures teams to establish robust testing protocols and sequencing while managing P&ID creation and analysis
- In overseeing the design, testing, and integration of fluid systems, prioritized safety, effectively managing risks associated with dangerous cryogenics and explosive oxidizers and fuels; notably, developed nearly the entirety of the FireX system by optimizing layout and pressure specifications to protect the engine bay and ensure clean integration with other hardware
- Pioneered a mobile GSE incorporating all the fluids onto one trailer, assembled the connections in an organized manner,
- Wrote a procedure and autosequence for LOx Fill/Drain, setup and stayed in the field during proof tests and cold flows.

**Systems for Augmenting Human Mechanics (SAHM) Lab** *Undergraduate Researcher, General Engineer* ----- Oct. 2023 – Jan. 2025

- Analyzed research documents, and produced an orthosis to assist patients with impaired mobility in walking, by using bands to redistribute work done by muscles in the rear of the leg to degenerated ones in the front, overall easing the swing phase.
- Set up and applied an electromyography (EMG) system to analyze neuromuscular signals, displaying raw signal strength and converting muscle activity into actionable data to control simulated or real prosthetics.

**Nuclear and Applied Robotics Group (NRG)** – Austin, TX ----- Jan. 2024 – May 2024

- In a team of 3, Developed a cutting-edge 6DOF Ceramic 3D Printer module, adaptable to any URD3 robotic arm.
- Owned the entire extruder assembly, prototyped complex housing and augur geometries to evenly lay high-viscosity clay.

**Relevant Course Projects, UNM** – Los Alamos, NM ----- Jan. 2022 – May 2023

- Designed, built, and wired a functional robotic tank featuring a high-powered 2-axis pneumatic cannon. Its abilities were put on display by the university, and I became an ambassador to local schools, encouraging students to join a robotics program.
  - Repeated the process with an off-road vehicle, while focusing on a robust spring-based suspension and differential 4WD
  - Developed end effectors for and programmed a FANUC standard industrial robot, enabling it to make, pour, and serve tea
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### COMMUNITY INVOLVEMENT

**Helping Hand Home** – Austin, TX ----- Jan. 2025 – Present

- Created, tested, sourced parts for, and performed fun interactive Science Demos to nurture students interest in STEM fields.
  - Forged valuable personal relationships with children (age 7-12) who require more attention that foster care can provide.
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### RELEVANT SKILLS

- **Design Software:** SolidWorks (including FEA), OnShape, SketchUp, Fusion360 (CAM), MakeraCAM, FDM & SLA 3D Printing
- **Robotic Systems:** Python, C++, ROS, General Electronics Layout and Assembly.
- **Certifications:** FANUC Industrial Robot Technician, Machine Shop, Programmable Logic Controllers, Safety Certifications (Fire, Cryogen, Compressed Gases, General Laboratory), Bilingual Seal for German Proficiency.

\*LinkedIn alternate link: <https://www.linkedin.com/in/longevan217/>

\*\*Portfolio website alternate link. View more of my projects here: <https://longevan217.wixsite.com/evan-long-portfolio>